

Playtester Telemetry Analysis

This notebook analyzes telemetry data collected from playtesters in order to understand player behavior and identify potential issues within the game level.

Select data file

data\playtest1_data\telemetry_dev.jsonl

Data Summary

- **Total records:** 11,077
- **Unique sessions:** 130
- **Time range:** 0.0s - 362.3s

Telemetry Events recieved per user

user_name str	count int64
clacsado2	4,873
koneillsalki	1,563
ltataryn2	1,289
spinto	1,208
cdaly3	825
mjankowski40	432
fireland2	409
adesjarlais31	371
rparedes2	77

9 rows, 1 column

Preview of the raw data

machine_id str	session_id str	user_name str	event_type str	frame int64	game_time float64	run_data object	server_timestam str
ITS1052719	ITS1052719_20260213_110914	rparedes2	position	12	3.013177394866943	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052719	ITS1052719_20260213_110914	rparedes2	position	19	4.766646862030029	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052719	ITS1052719_20260213_110914	rparedes2	position	20	5.010434150695801	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052719	ITS1052719_20260213_110914	rparedes2	position	21	5.264326572418213	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052719	ITS1052719_20260213_110914	rparedes2	position	24	6.013036727905273	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	74	18.5015550674438477	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	70	17.50233268737793	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	75	18.750635147094727	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	71	17.757373809814453	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000
ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	73	18.258941650390625	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	2026-02-13T11:14:00.000000

10,819 rows, 16 columns

Data Cleaning Pipeline

Removing outliers and invalid data points.

Showcase of what raw data graphed looks like



Drop data from people who have passed the end of the level

	machine_id	session_id	user_name	event_type	frame	game_time	run_data	server_time
int64	str	str	str	str	int64	float64	object	str
0	ITS1052719	ITS1052719_20260213_110914	rparesd2	position	12	3.013177394866943	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
1	ITS1052719	ITS1052719_20260213_110914	rparesd2	position	19	4.766646862030029	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
2	ITS1052719	ITS1052719_20260213_110914	rparesd2	position	20	5.010434150695801	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
3	ITS1052719	ITS1052719_20260213_110914	rparesd2	position	21	5.264326572418213	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
4	ITS1052719	ITS1052719_20260213_110914	rparesd2	position	24	6.013036727905273	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
5	ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	74	18.501550674438477	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
6	ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	70	17.50233268737793	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
7	ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	75	18.750635147094727	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
8	ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	71	17.757373809814453	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13
9	ITS1052703	ITS1052703_20260213_110901	ltataryn2	position	73	18.258941650390625	{'run_id': '', 'run_start_time': 0, 'run_end_time': 0, 'run...	2026-02-13

10,190 rows, 16 columns

Sort data

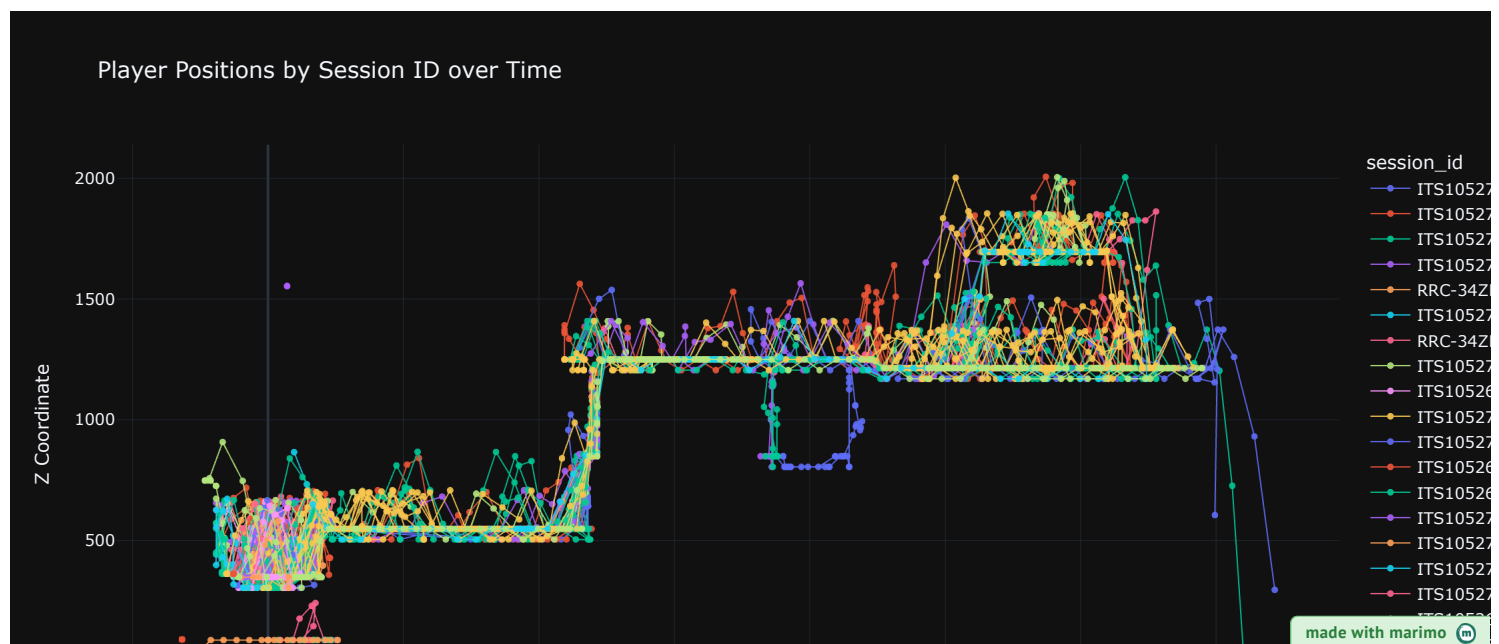
	machine_id	session_id	user_name	event_type	frame	game_time	run_data	status
int64	str	str	str	str	int64	float64	object	str
3,950	ITS1052698	ITS1052698_20260213_113033	fireland2	position	1	0.40833342075347906	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,961	ITS1052698	ITS1052698_20260213_113033	fireland2	position	2	0.8083333969116211	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,962	ITS1052698	ITS1052698_20260213_113033	fireland2	position	3	0.8083333969116211	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,965	ITS1052698	ITS1052698_20260213_113033	fireland2	position	4	1.082142829895019	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,968	ITS1052698	ITS1052698_20260213_113033	fireland2	position	5	1.264379501342773	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,971	ITS1052698	ITS1052698_20260213_113033	fireland2	position	6	1.566142082214355	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,974	ITS1052698	ITS1052698_20260213_113033	fireland2	position	7	1.7800579071044922	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,977	ITS1052698	ITS1052698_20260213_113033	fireland2	position	8	2.019359588623047	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,980	ITS1052698	ITS1052698_20260213_113033	fireland2	position	9	2.287259101867676	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20
3,983	ITS1052698	ITS1052698_20260213_113033	fireland2	position	10	2.52285885810852	{'run_id':"", 'run_start_time':0, 'run_end_time':0, 'run_tot...	20

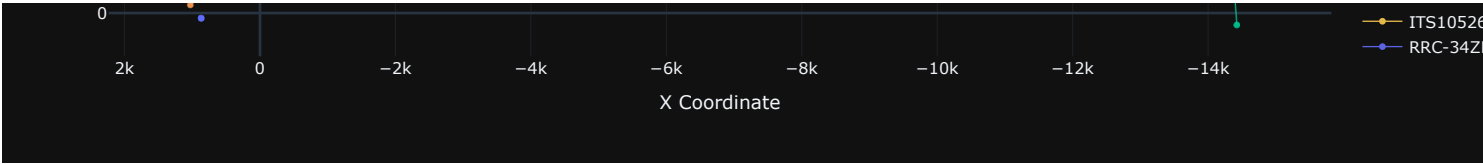
10,190 rows, 16 columns

Ensure all data is in correct time sequence

- Time sequence violations: 0

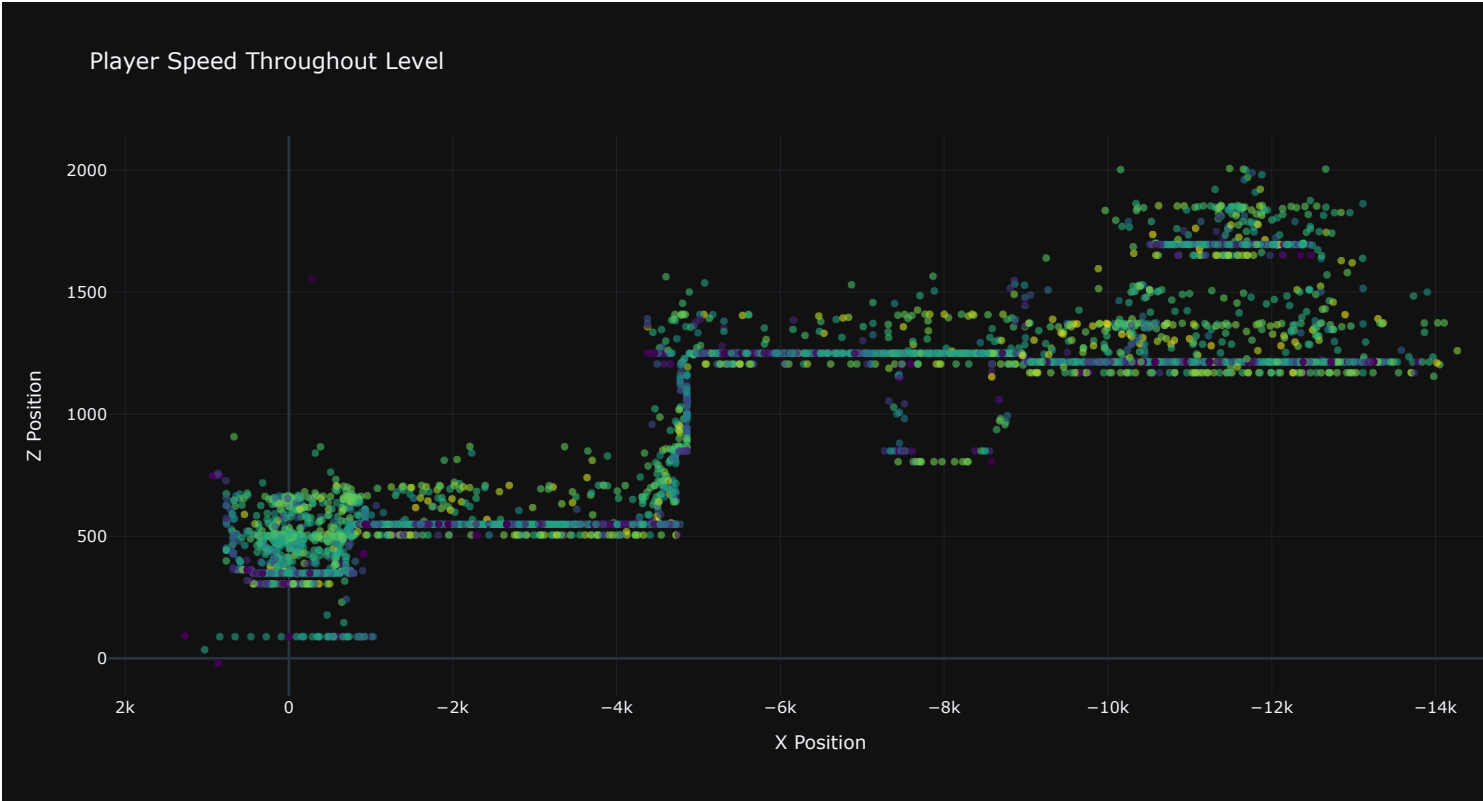
Player Pathing visualizaton





Player Velocity Analysis

Shows where players speed up or slow down



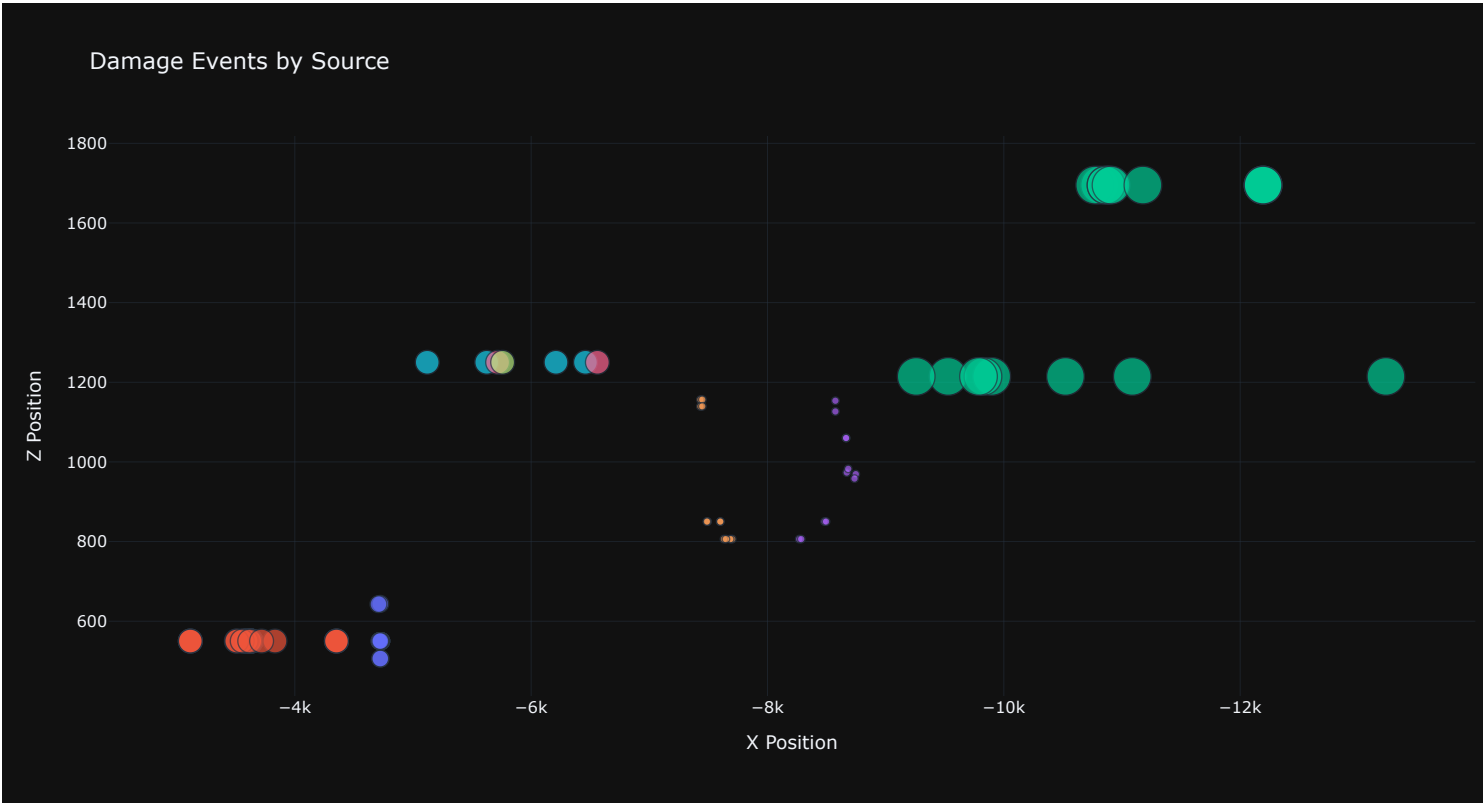
Damage and Death Analysis

Damage & Death Analysis

Using the new telemetry fields: event_type, damage, health_before, health_after, damage_source, run_data.

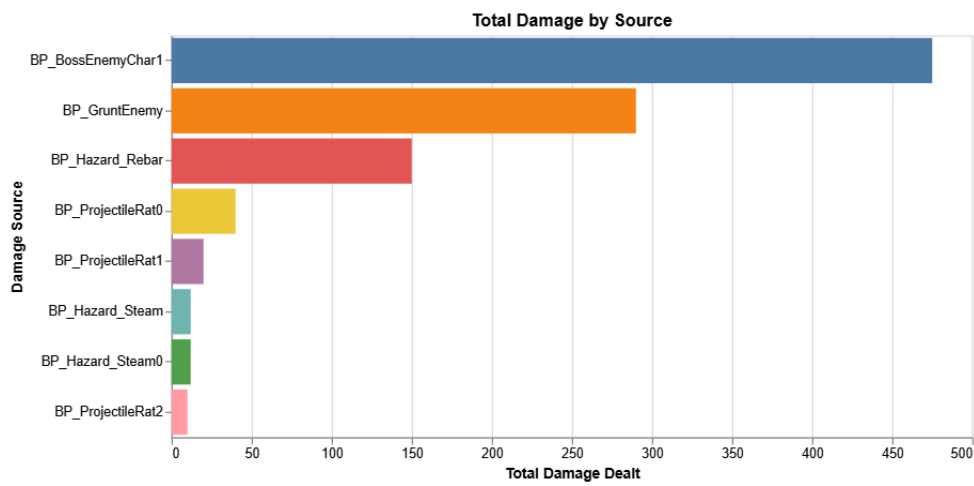
Damage Events by Source

Each point is a damage event. Color coding is representative of source, while size represents the amount of damage taken.



Damage by Source

Which enemy or hazard is dealing the most damage overall?



Player Health Over Time

This shows how the player's health decreases over the duration of their run.

IMPORTANT NOTE: This does NOT correspond to position in the level, this is over time.

Health Over Time by Session

